

Ioannis Asmanis

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EDUCATION

University of Pennsylvania

Aug 2022 - *present*

School of Engineering and Applied Sciences, GRASP Lab

PhD Student, Current GPA: 3.98

Research interests: Computer Vision, Neuromorphic Computing, Robotics

Advisor: Prof. Kostas Daniilidis

National Technical University of Athens

Oct 2014 - Jun 2021

School of Electrical and Computer Engineering

Integrated MEng Diploma, Final GPA: 9.1/10 (Excellent)

Specializations: Signals, Robotics & Control, Software, Computer Systems, Mathematics

Thesis: “*Combining semantics with unified geometric representations for indoor SLAM*”, grade 10/10

Supervisor: Prof. Petros Maragos

EXPERIENCE

Raytheon Technologies Corporation

May 2024 - Jan 2025

Research Intern

RTX Research Center, Hartford, CT

Energy-efficient neuromorphic language models.

- literature review for Spiking Neural Networks in LLM applications
- designed, coded, and trained novel architecture
- benchmarked inference energy consumption with positive results

Hellenic Army Research and Technology Center

Jul 2021 - Jul 2022

Private, Junior Robotics Engineer

Hellenic Armed Forces, Athens

Autonomous patrol and surveillance rover and other projects.

- modeling and software development for an autonomous wheeled platform for outdoor environments
- full-stack development of distributed sensing weather platform
- contributed to various vision & micro-controller projects

Intelligent Robotics & Automation Lab (IRAL)

Jan 2021 - Jun 2021

Research Assistant, ATHENA Research Center scholar

NTUA, Athens

Healthcare assistive robotics project: “iWalk” robotic walker

- designed & developed DNN models for real-time gesture recognition
- refactored legacy code and managed datasets
- demonstrated the technology in a field test at the Diapylis Rehabilitation Hospital

European Space Agency (ESA)

Jul 2019 - Jan 2020

Planetary Robotics Laboratory Intern

ESTEC, Noordwijk, Netherlands

Robotic vision for autonomous planetary exploration rovers

- FPGA-accelerated prototype Visual Odometry algorithm verification and integration
- tested on rovers in Mars Analogue terrain, interfacing with real mission control software
- contributed a stereo image rectification package and participated in multiple field tests

PUBLICATIONS

“A Scalable, Causal, and Energy Efficient Framework for Neural Decoding with Spiking Neural Networks”

I. Asmanis*, G. Mentzelopoulos*, K. Kording, E. Dyer, K. Daniilidis, F. Vitale - NIPS 2025

“Motion-prior Contrast Maximization for Dense Continuous-Time Motion Estimation”

F. Hamann*, Z. Wang, **I. Asmanis**, K. Chaney, G. Gallego, K. Daniilidis - ECCV 2024

“A Semantic Enhancement of Unified Geometric Representations for Improving Indoor Visual SLAM”

I. Asmanis*, P. Mermigkas, G. Chalvatzaki, J. Peters, P. Maragos - UR 2022

TECHNICAL SKILLS

Programming Languages Python, C/C++, Java, Matlab, SQL/MySQL, Verilog, C#, Rust

Frameworks & Tools PyTorch, NumPy, Jupyter, OpenCV, ROS/ROS2, Rock, PCL, Eigen, Arduino, RaspberryPi, Linux, LaTeX, Office

AWARDS AND SCHOLARSHIPS

Scholarships

Bodossaki Foundation	Partial scholarship for graduate studies	2022 - 2027
ATHENA Research Center	Scholarship for RA at IRAL, NTUA	2021
Google	Associate Android Developer Scholarship	2017
DAAD	Scholarship for MASCOR ROS summer school	2016

University matriculation awards

NTUA	Award for ranking 1 st in admissions	2014
Eurobank	“Great moment for education” prize	2014
Cosmote	Merit scholarship for academic achievement	2014

Competitions

45 th IPhO, Kazakhstan	Honorable mention	2014
National Physics Competition	Finalist (top-15 nationwide)	2012 - 2014
Greek Mathematical Olympiad	Bronze medals	2010, 2011

ASSISTANSHIPS AND VOLUNTEERING

TA: CIS 5800 Fall 2023
Head Graduate Teaching Assistant UPenn, PA

Lead of TAs for Computer Vision graduate course.

TA: Computer programming Oct 2016 - Jun 2017
Undergraduate TA NTUA, Athens

Assisted in the organization and instruction of C/C++ to first-year ECE undergraduate students.

Conference Reviewer: NIPS 2025, ECCV 2024, IROS 2021, ICRA 2020

Other volunteering

- **ESA/ESTEC:** Open Day volunteer *Oct 2019*
- **Athens Science Festival:** Science Ambassador & School Guide, Technopolis, *2015 - 2017*

LANGUAGES

English: fluent, **German:** intermediate, **Greek:** first language